

Yuxiao Wen

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EDUCATION

NEW YORK UNIVERSITY

New York, NY

Ph.D. in Computer Science

Expected 2027

Co-advised by Yanjun Han and Zhengyuan Zhou

GPA: 3.95/4.0

recipient of MacCracken fellowship, NYU AI Safety fellowship

M.Phil. in Computer Science

Jan 2026

B.A.s in Honors Mathematics, Computer Science

May 2021

GPA: 3.88/4.0

department's outstanding award (2 per year), magna cum laude (top 15%), Dean's List

RESEARCH INTERESTS

Reinforcement learning, Contextual bandits, Online learning, Statistical inference, Operations research.

PREPRINT

- [19] Y. Wen. **Fast A/B/n Testing: Exact Multi-Policy Comparison via Tree-Coupled Feedback Sharing**, submitted to *International Conference on Learning Representations (ICLR)*, 2026. (arXiv: 2608.12831)
- [18] Y. Wen, Y. Han, Z. Zhou. **Joint Value Estimation and Bidding in Repeated First-price Auctions**, submitted to *Management Science*, 2026. (POMS-HK 2026 Best Student Paper Finalist, POMS-China 2026 Best Student Paper Finalist, arXiv: 2502.17292)
- [17] Y. Wen, Y. Pang, Y. Han, Z. Deng. **Trustworthy On-policy Distillation under Operational Constraints**, *Working draft*, 2026.
- [16] Q. Bai^(*αβ), Y. Han^(*αβ), Y. Wen^(*αβ), W. Xu, Z. Zhou. **Functional Critics Are Essential for Actor-Critic: From Off-Policy Stability to Efficient Exploration**, submitted to *International Conference on Learning Representations (ICLR)*, 2026. (arXiv: 2509.22964).
- [15] K. Zhang, L. Zhang, Y. Pang, Y. Wen, Z. Deng, Y. Han. **Constrained Gradient Equilibrium: Approachability under Locally Constrained Adversaries**, submitted to *International Conference on Artificial Intelligence and Statistics (AISTATS)*, 2026.
- [14] Y. Wen, Y. Han, Z. Zhou. **Causal Inference Bandits for Treatment Allocations**. *Working draft*, 2026.
- [13] ^(αβ)C. Chen, W. Lu, X. Lu, Y. Wen, J. Yang, Z. Zhou. **Learning to Activate and Scale: Price-based Semi-bandit Algorithms for Resource Allocation**, *Working draft*, 2026.
- [12] Y. Wen, J. Huang, W. Zhou, Z. Zhou. **Online Inventory Control with Context-Aware Demand Distributions**, reject&resubmit at *Manufacturing & Service Operations Management (M&SOM)*, 2026.
- [11] Z. Hu^(*αβ), Y. Wen^(*αβ), Y. Yao, J. Zhang, Z. Zhou. **Learning to Bid with Unknown Private Values in Budget-Constrained First-Price Auctions**, preprint, 2026. (arXiv: 2605.09448)

PUBLICATIONS

- [10] Y. Wen, Z. Hu, Y. Han, Y. Yao, Z. Zhou. **The (Marginal) Value of a Search Ad: An Online Causal Framework for Repeated Second-price Auctions**, *International Conference on Machine Learning (ICML)*, 2026.
- [9] Y. Wen, Y. Han, Z. Zhou. **Optimal Arm Elimination Algorithms for Combinatorial Bandits**, *Conference on Artificial Intelligence and Statistics (AISTATS)*, 2026.
- [8] Y. Han^(*αβ), X. Xu^(*αβ), Y. Wen, Y. Han, I. Lobel, Z. Zhou. **Semi-Parametric Contextual Pricing with General Smoothness**, *International Conference on Learning Representations (ICLR)*, 2026.

- [7] Y. Wen. **Adversarial Combinatorial Semi-bandits with Graph Feedback**, *International Conference on Machine Learning (ICML)*, 2025.
- [6] D. Feijer et al. **Calibrated Recommendations with Contextual Bandits**, *ACM Recommender Systems (RecSys) Workshop*, 2025.
- [5] A. Jacot, S.H. Choi, Y. Wen. **How DNNs break the Curse of Dimensionality: Compositionality and Symmetry Learning**, *International Conference on Learning Representations (ICLR)*, 2025.
- [4] Y. Wen, Y. Han, Z. Zhou. **Stochastic contextual bandits with graph feedback: from independence number to MAS number**, *Neural Information Processing Systems (NeurIPS)*, 2024.
- [3] Y. Wen, A. Jacot. **Which Frequencies do CNNs Need? Emergent Bottleneck Structure in Feature Learning**, *International Conference on Machine Learning (ICML)*, 2024.
- [2] Y. Wen, E. Vanden-Eijnden, B. Peherstorfer. **Coupling parameter and particle dynamics for adaptive sampling in Neural Galerkin schemes**, *Physica D: Nonlinear Phenomena*, 2024.
- [1] W.I.T. Uy, Y. Wang, Y. Wen, and B. Peherstorfer. **Active operator inference for learning low-dimensional dynamical-system models from noisy data**, *SIAM Journal on Scientific Computing*, 2023.

TEACHING

- [3] DS-GA 3001: Applied Statistics, Fall 2024
- [2] MATH-UA 133: Math for Econ III, Spring 2024
- [1] DS-GA 3001: Applied Statistics, Fall 2023

AWARD

- [5] Best Student Paper Third Prize, POMS-China, 2026
- [4] Best Student Paper Finalist, POMS-HK, 2026
- [3] Silver Reviewer Award, ICML, 2026
- [2] NYU Dean's Conference Fund, 2025
- [1] NYU CS Department Outstanding Performance Award, 2021 (2 per year)

TALKS

- [3] POMS-China, *Joint Value Estimation and Bidding in Repeated First-price Auctions*, Xi'an, China, 2026.
- [2] POMS-HK, *Joint Value Estimation and Bidding in Repeated First-price Auctions*, Hong Kong, China, 2026.
- [2] DAO Seminar at National University of Singapore, *Joint Value Estimation and Bidding in Repeated First-price Auctions*, Singapore, 2025.
- [1] SIAM Conference on Computational Science and Engineering, *Adaptive Sampling for Efficiently Training Models of Nonlinear Latent Dynamics*, Amsterdam, the Netherlands, 2023.

WORK EXPERIENCE

Amazon Web Services, Inc.

Seattle, WA

Applied Scientist Intern

May 2025 - Aug 2025

- Proposed learning algorithms for setting floor prices in online advertising with >1M request per minute.
- Adapted RL algorithms for dynamic floor pricing that achieve >3% increase in publisher's revenue.

Spotify USA, Inc.

New York, NY

Machine Learning Engineer Intern

June 2024 - Aug 2024

- Implemented a sophisticated exploration improving data quality and user experience for ~5M users daily.
- Adopted a contextual bandit algorithm giving ~23% relative increase from baseline in user click rate.

PROFESSIONAL SERVICE

Conference Reviewer: ICML 2024-2026, NeurIPS 2024-2025, ICLR 2025-2027, AISTATS 2026

Journal Reviewer: Journal of Machine Learning Research, Management Science, Math of Operations Research